

**Multi-agent
cooperation**

2. AnalysisAgent

Dynamic risk analysis
Initial tool generation



3. FusionAgent

Fast/slow tool retrieval
Fuse existing tools with new



4. ExecutorAgent

Sandbox executing user
queries and agent actions



5. AuditAgent

Audit tools;
Judge results; Refine tools



Lifelong Evolution Loop

Legend



Initial tool



Dynamic risk



Fused tool



Audited tool



Risky tool

Input



User queries



Agent actions



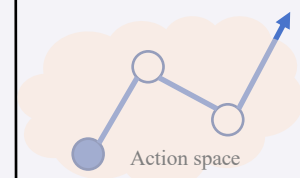
Risk library



Tool library

1. SimulationAgent

Predict action



Predict agent actions based on query

1 Generation



Initial tool
generation



Dynamic
find risks

2 Reuse



Retrieve tools
similar to the new



Tool fusion

Risk library

Update library
(Add new risk category)

Evolve tool
(Fuse new with existing)

Tool library

Update library
(Add refined tools)

Update library
(Add audited tools)

Audit tool
(Evoled, safety aspect)

3 Audit

Pass

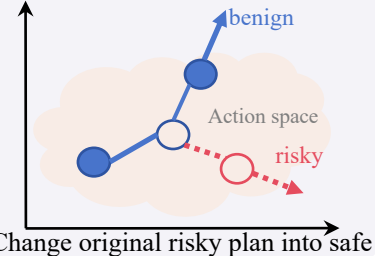
Fail

4 Refinement

Refine tools based
on audit result

6. RevisionAgent

if **risky**, refine action



Change original risky plan into safe

Output



Judge result



Refined action



Risk library



Tool library